Lesson 02 Demo 04

Creating a React Application Using Conditional Rendering

Objective: To build a simple React application using the Vite framework that demonstrates conditional rendering techniques, enabling dynamic UI updates based on user interaction

Tools required: Visual Studio Code and Node.js

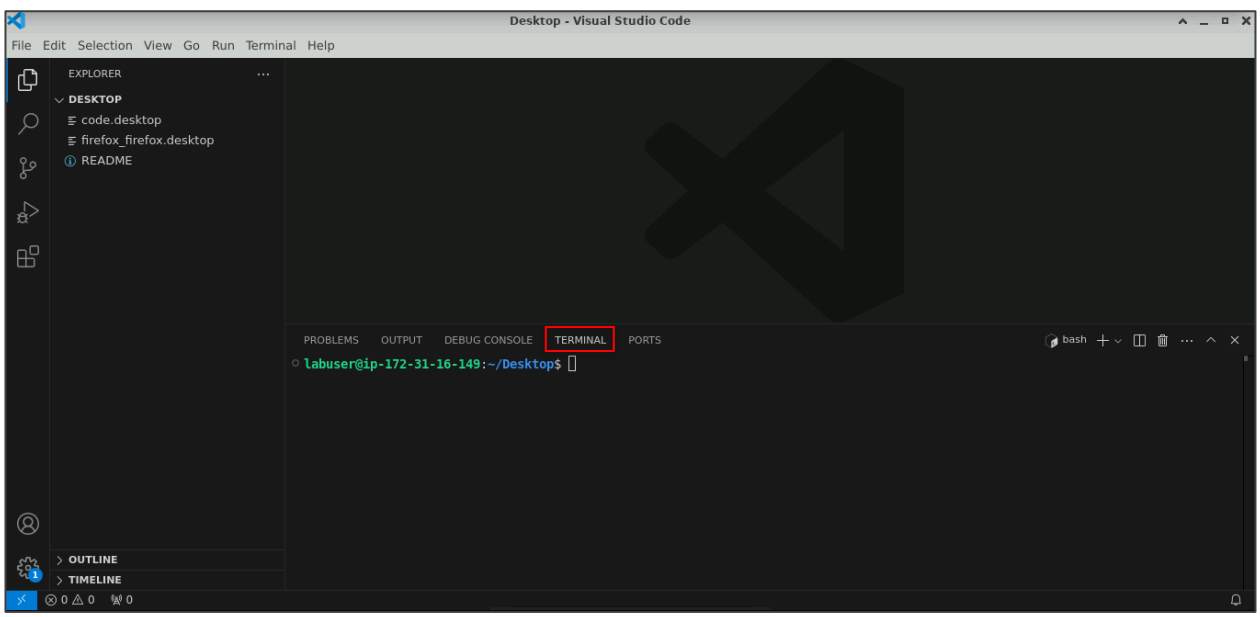
Prerequisites: None

Steps to be followed:

1. Set up a new React project using Vite
2. Create a React component with conditional rendering
3. Import and use the component
4. Run and test the application

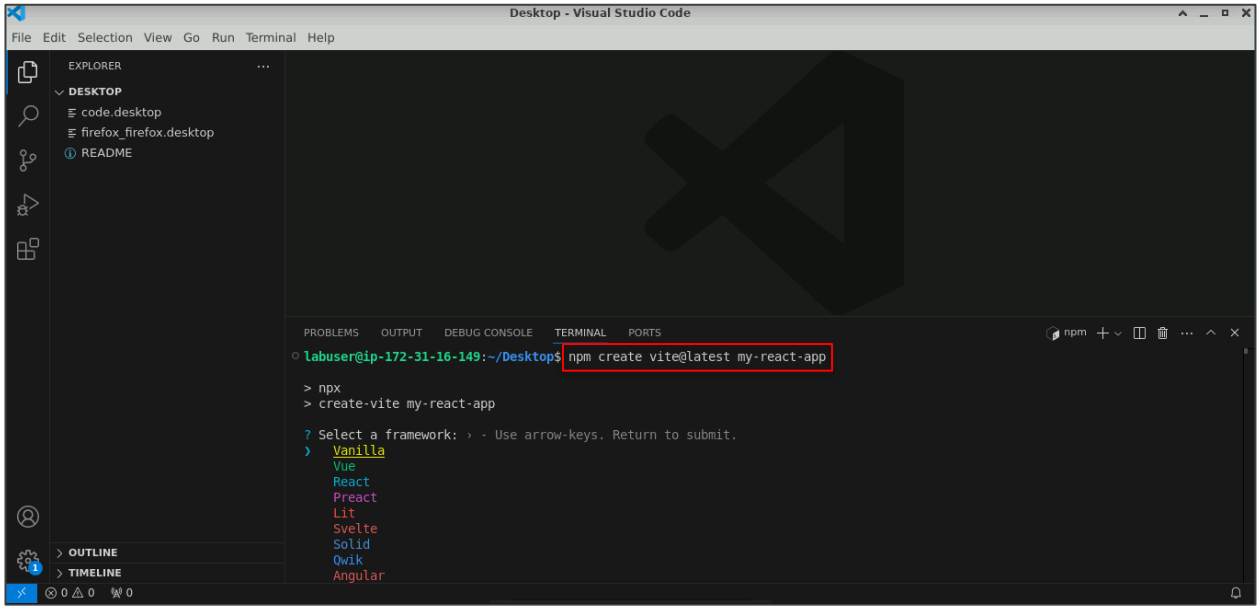
Step 1: Set up a new React project using Vite

1.1 Navigate to VS Code and open the terminal

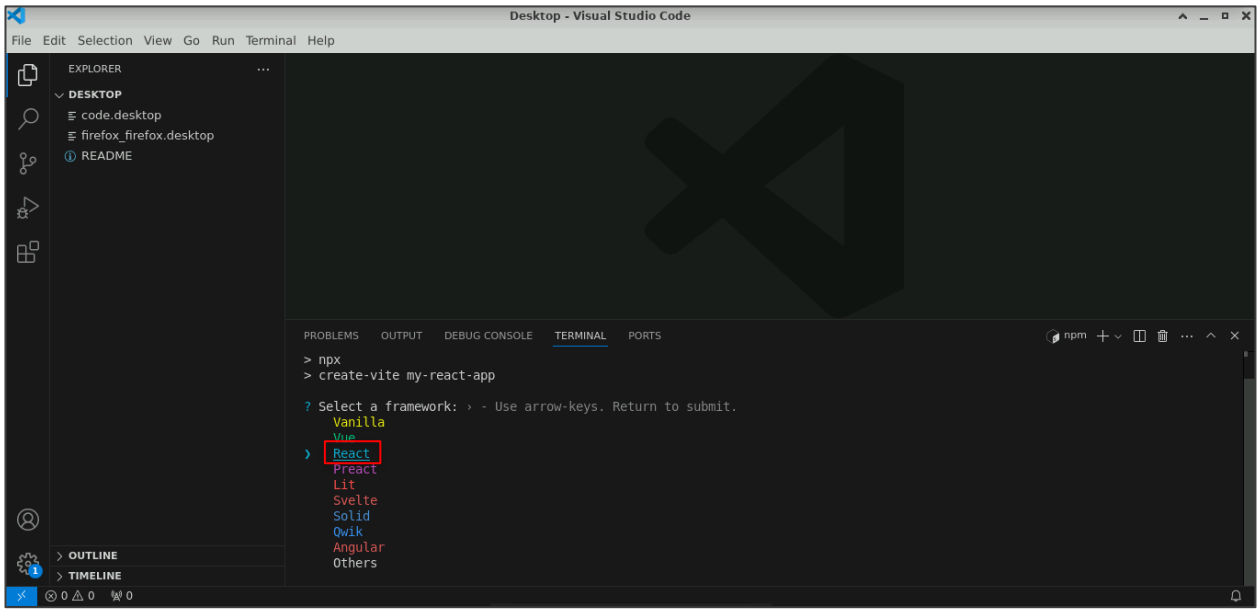


1.2 Run the following command to create a new React project using Vite:

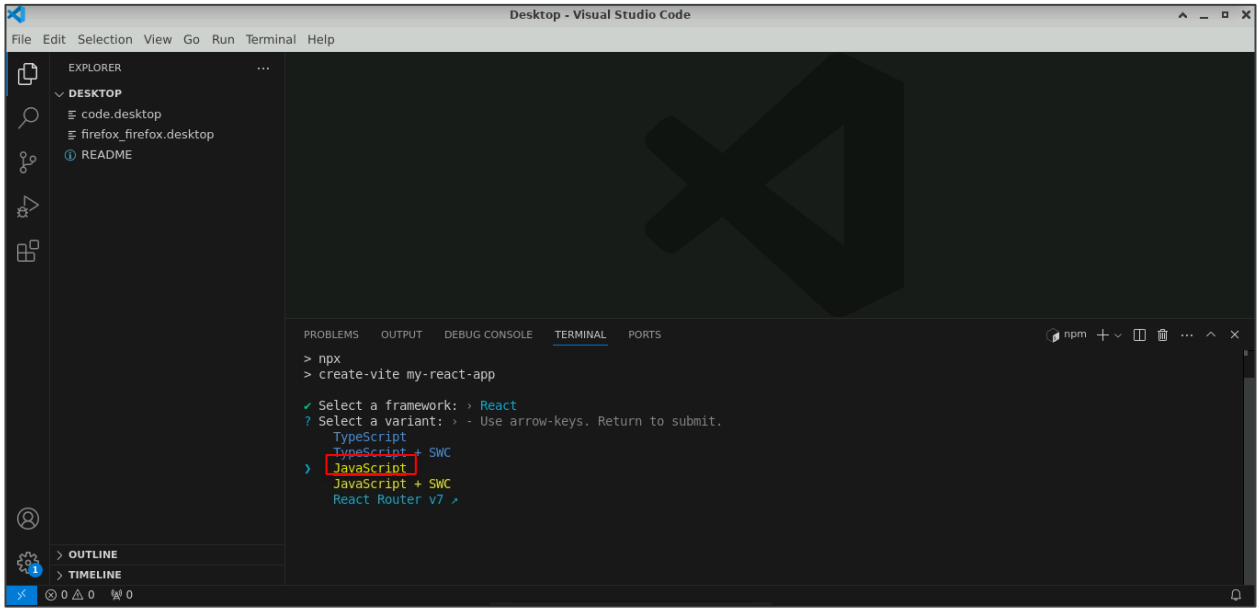
npm create vite@latest my-react-app



1.3 Select **React** and press enter

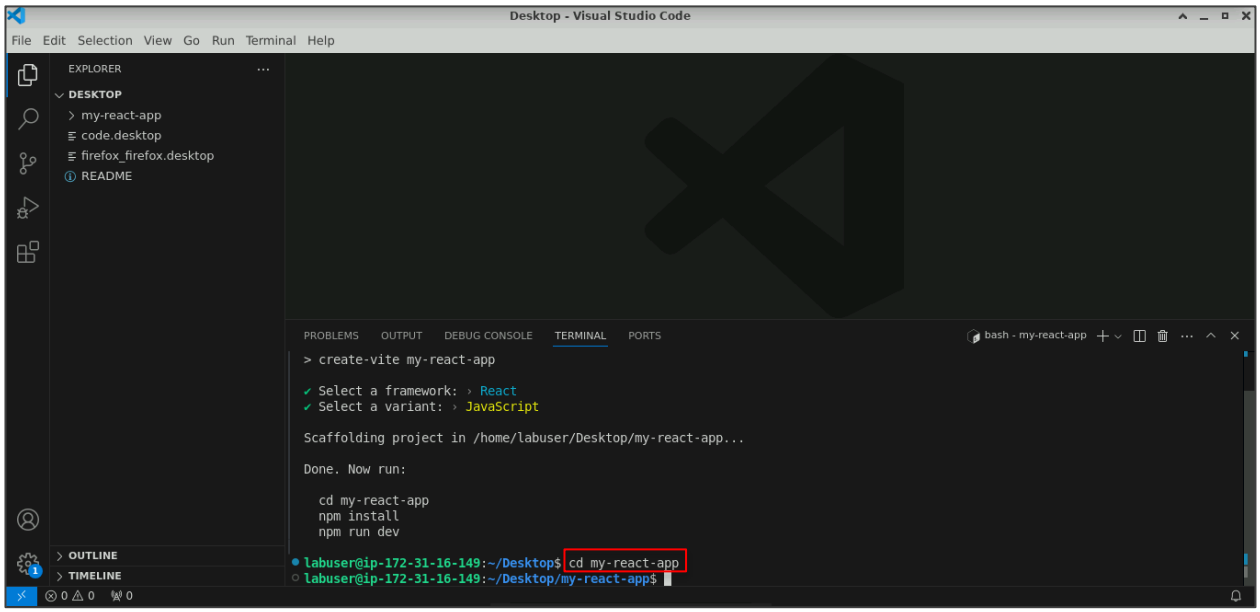


1.4 Select **JavaScript** and press enter



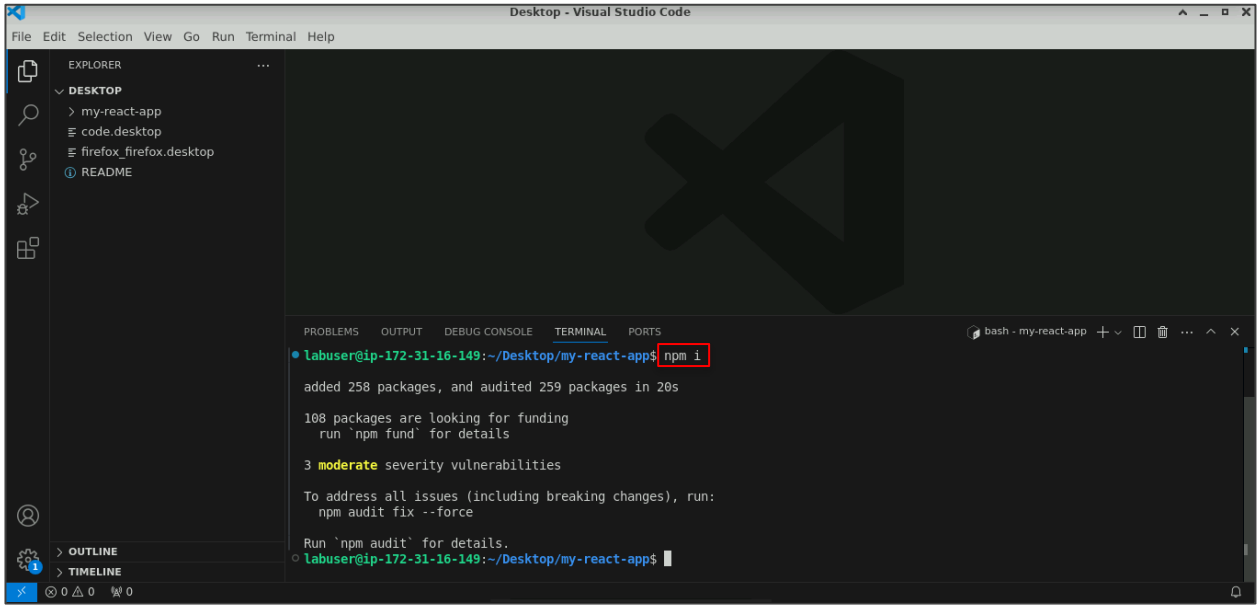
1.5 Run the following command to navigate to the project folder:

cd my-react-app



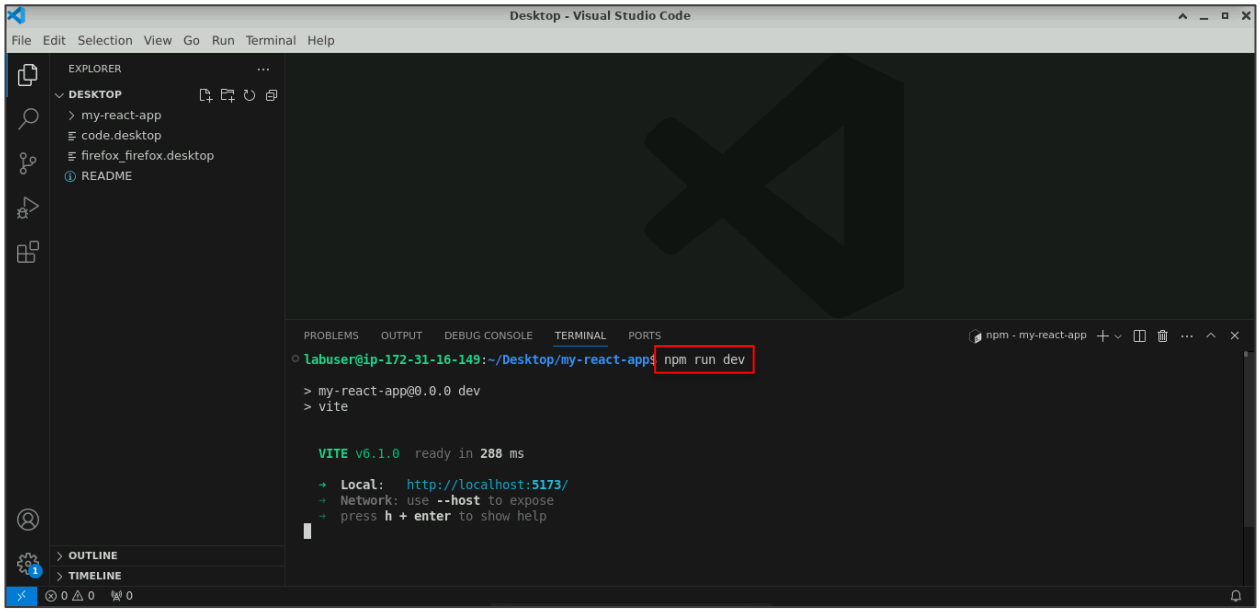
1.6 Run the following command to install dependencies:

npm i



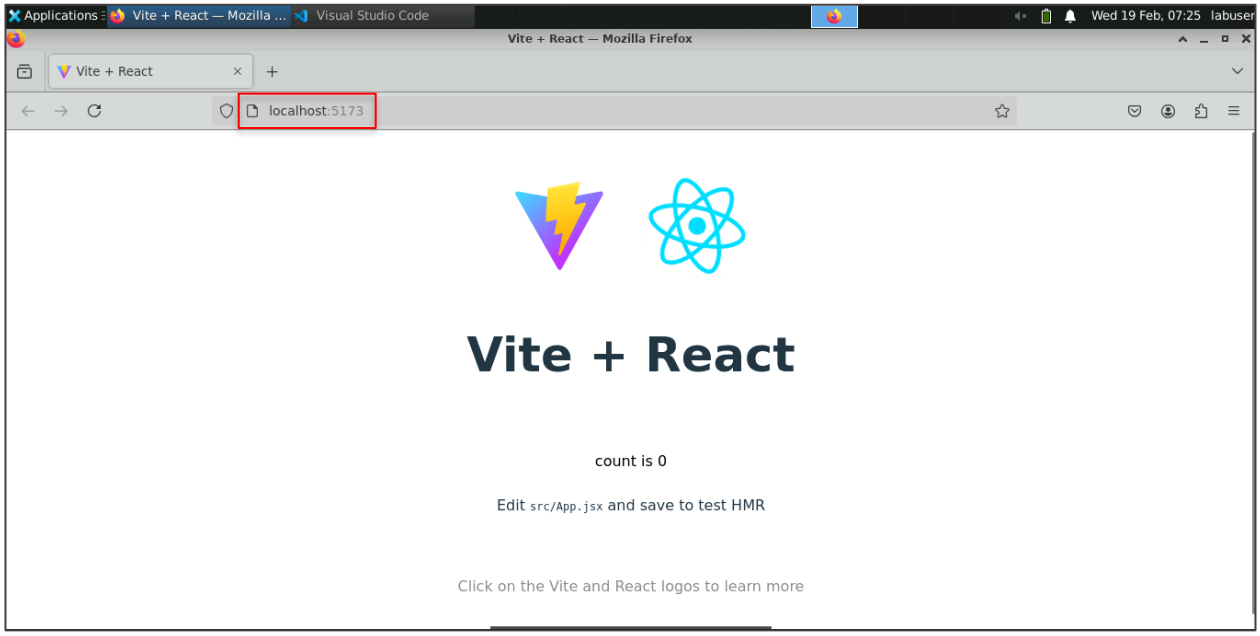
1.7 Run the following command to start the development server:

npm run dev



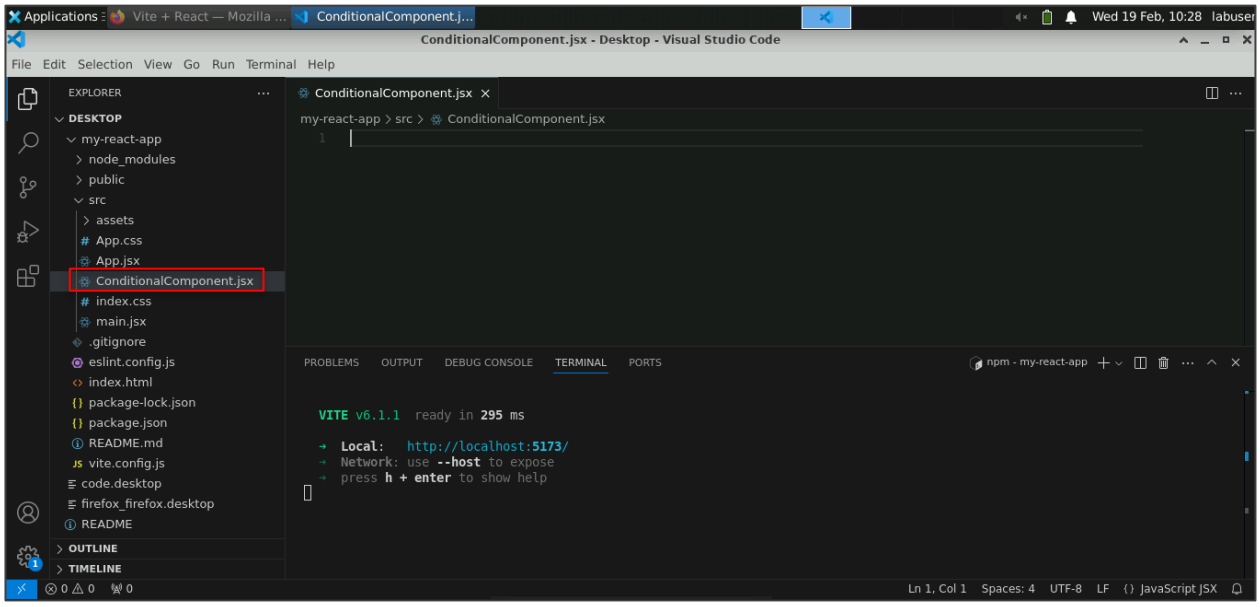
1.8 Navigate to the web browser and paste the link below:

http://localhost:5173



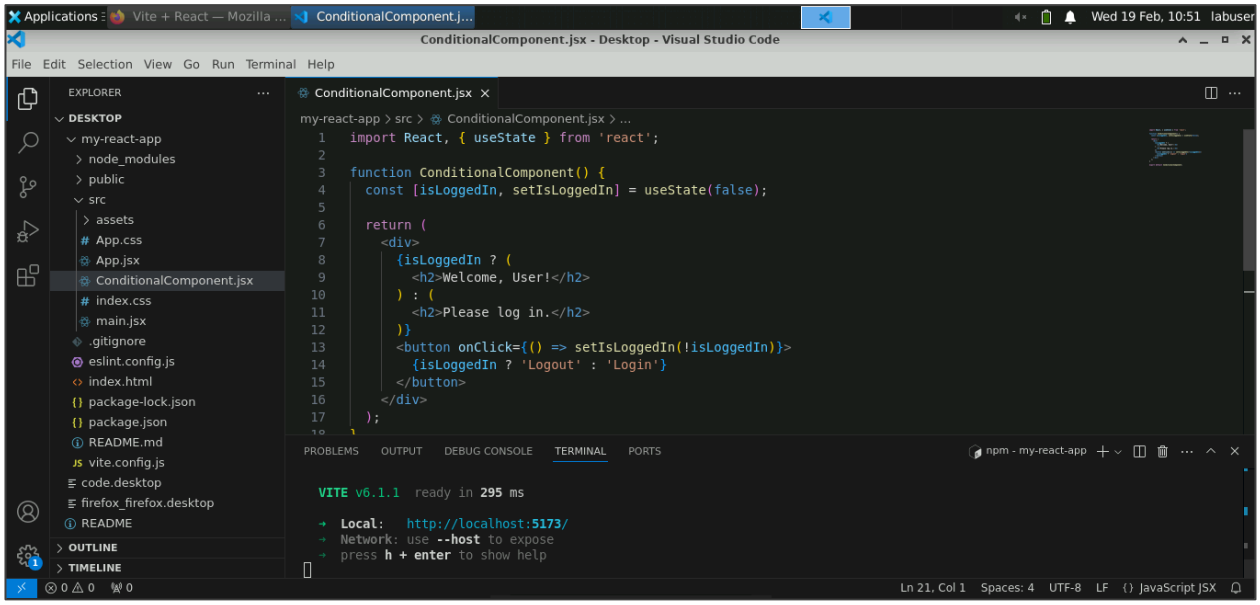
Step 2: Create a React component with conditional rendering

2.1 Inside the **src** folder, create a new file named **ConditionalComponent.jsx**



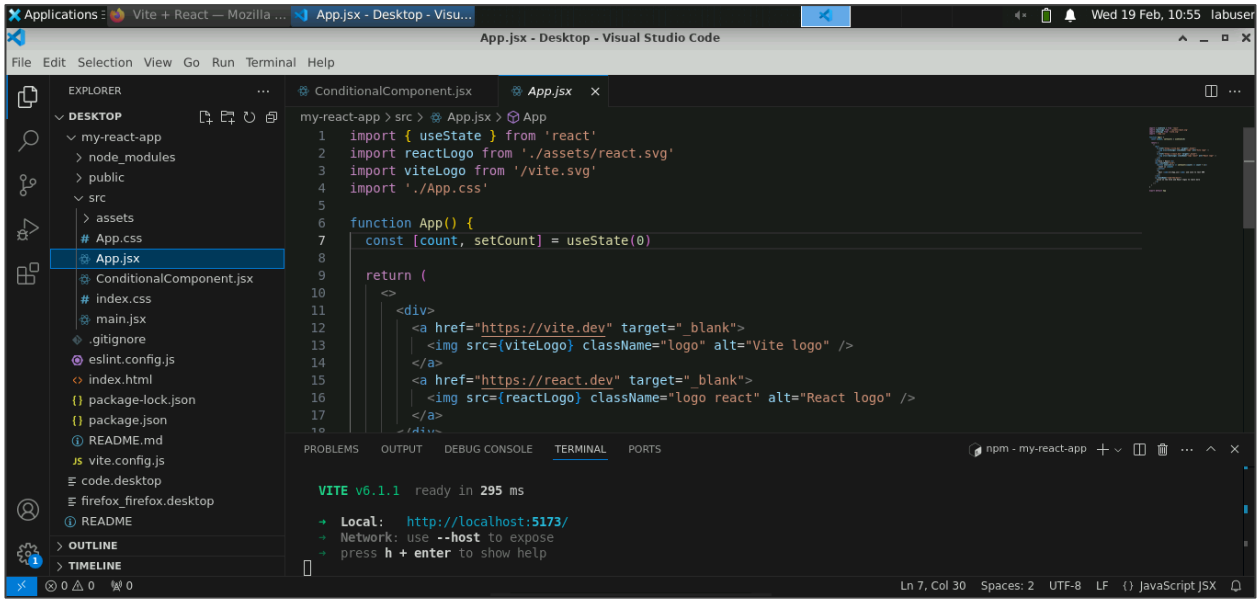
2.2 Add the following code to **ConditionalComponent.jsx:**

```
import React, { useState } from 'react';

function ConditionalComponent() {
  const [isLoggedIn, setIsLoggedIn] = useState(false);

  return (
    <div>
      {isLoggedIn ? (
        <h2>Welcome, User!</h2>
      ) : (
        <h2>Please log in.</h2>
      )}
      <button onClick={() => setIsLoggedIn(!isLoggedIn)}>
        {isLoggedIn ? 'Logout' : 'Login'}
      </button>
    </div>
  );
}

export default ConditionalComponent;
```



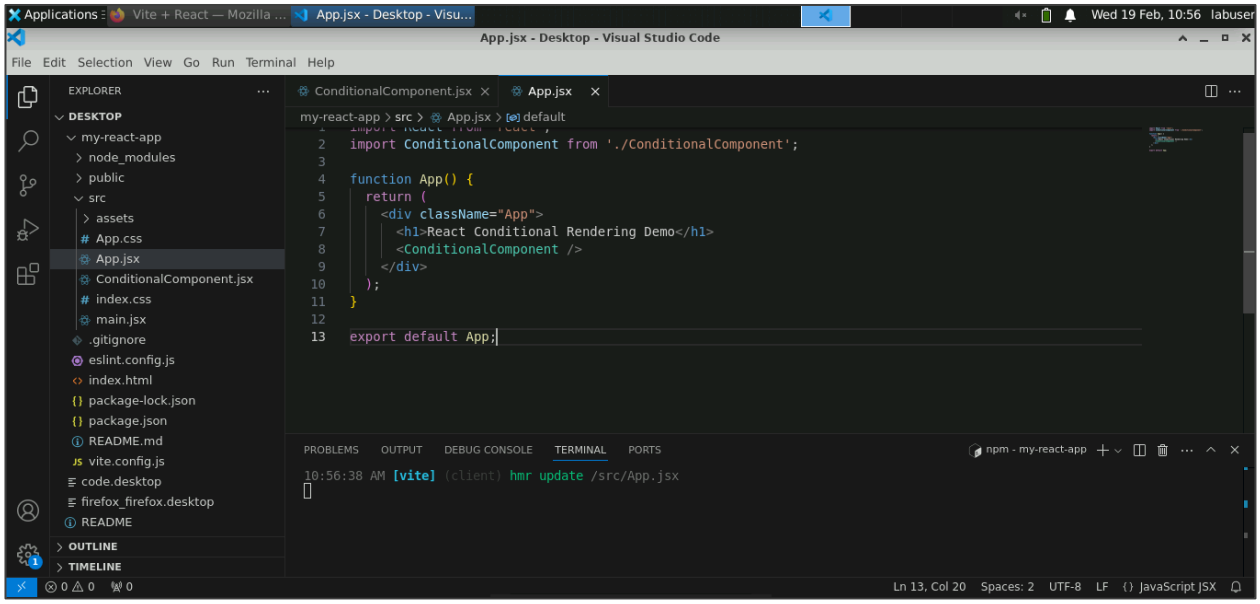
Step 3: Import and use the component

3.1 Open the **App.jsx** file inside the **src** folder



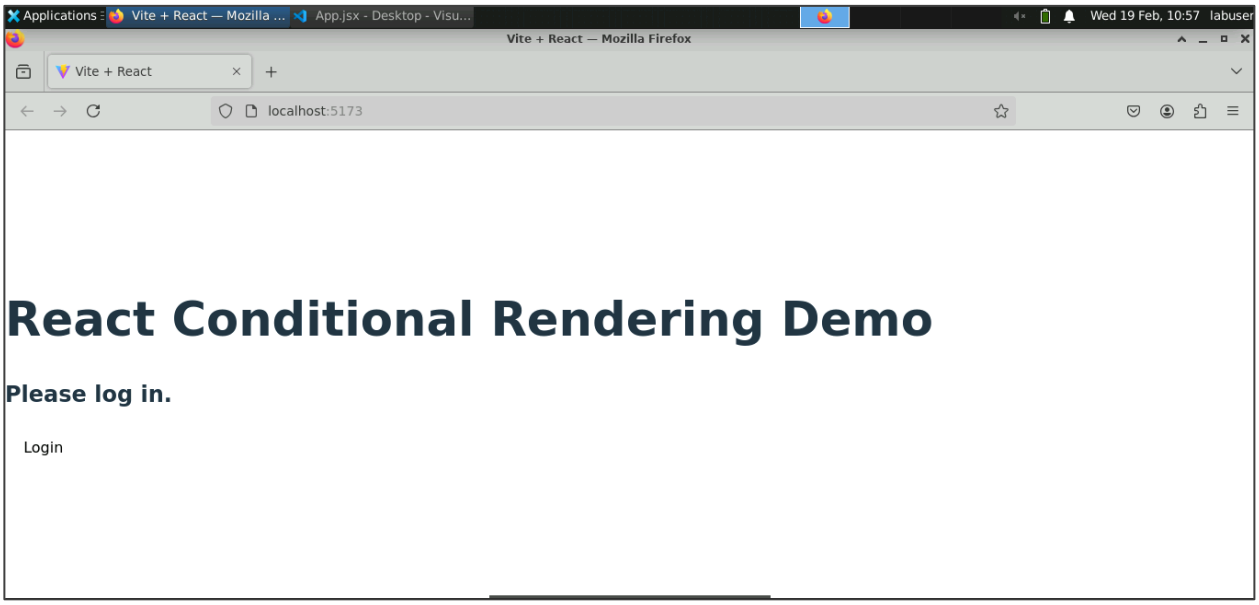
3.2 Modify the file to include the new component:

```
import React from 'react';
import ConditionalComponent from './ConditionalComponent';

function App() {
  return (
    <div className="App">
      <h1>React Conditional Rendering Demo</h1>
      <ConditionalComponent />
    </div>
  );
}

export default App;
```

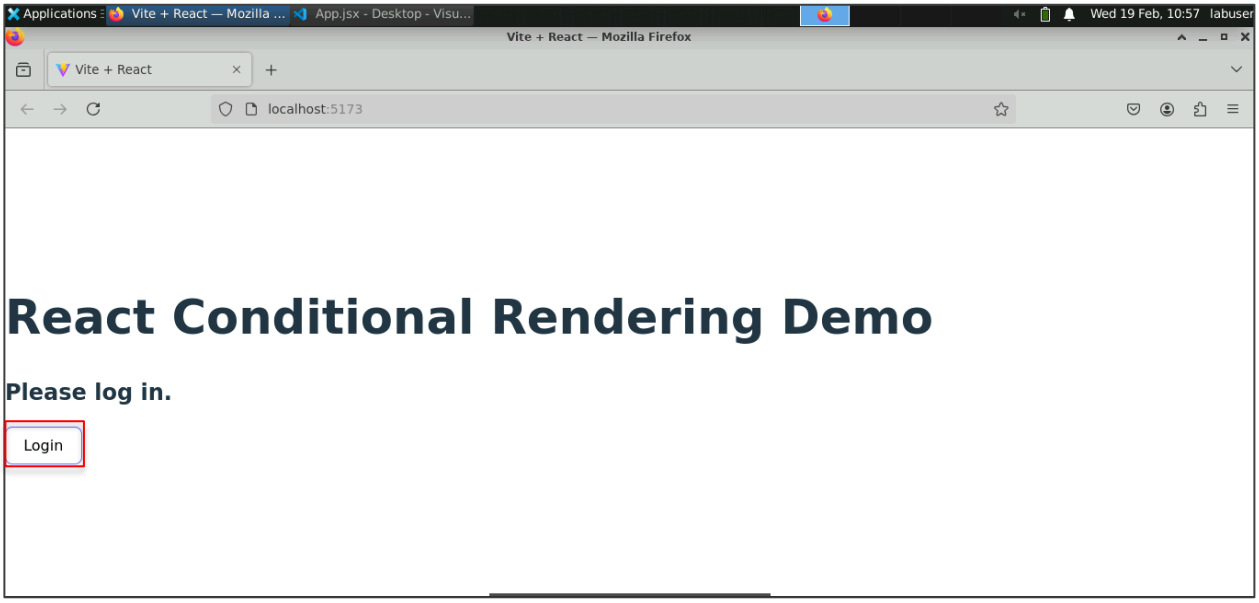


Step 4: Run and test the application

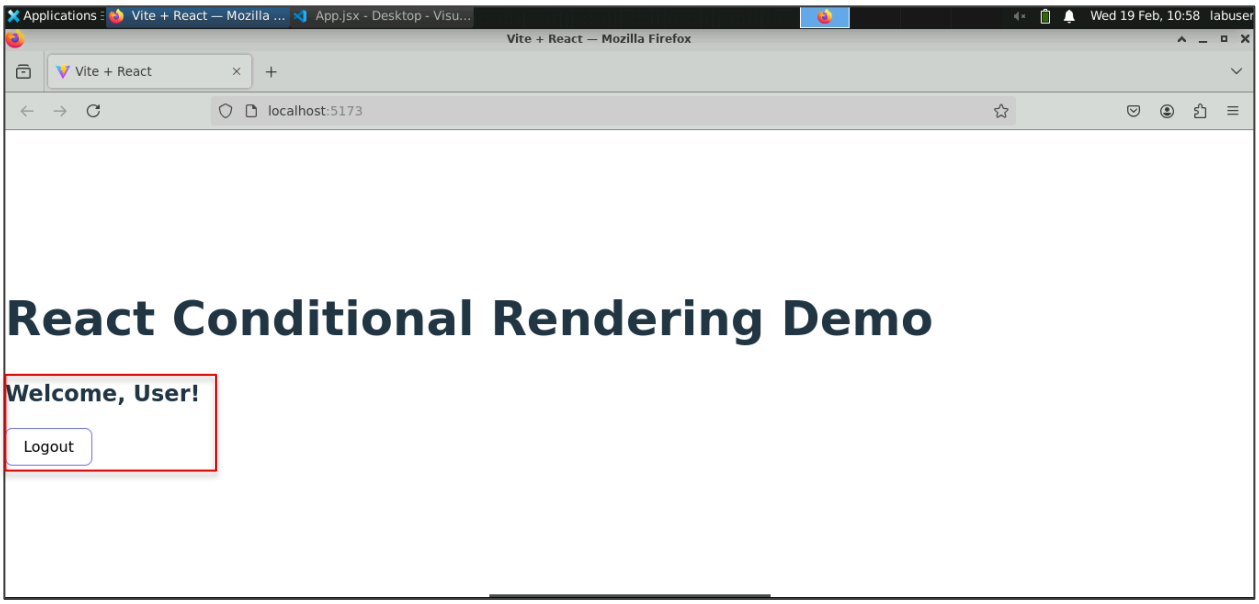
4.1 Open the localhost URL in a browser



4.2 Click **Login**



Now, you can view the change in content when the Login button is clicked.



By following the above steps, you have successfully created a simple React application using the Vite framework that demonstrates conditional rendering. This application dynamically updates the UI based on user interaction, showcasing how React handles state changes efficiently.